

Ali Raza

AI/ML Engineer — Convolutional Neural Networks — Deep Learning — NLP
Islamabad, Pakistan

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Summary

Results-driven AI/ML Engineer with hands-on experience developing and deploying machine learning and deep learning solutions using Python, Scikit-learn, TensorFlow, and PyTorch. Skilled in building end-to-end AI applications including sentiment analysis, fake news detection, customer segmentation, credit risk analysis, and image recognition systems. Experienced in data preprocessing, feature engineering, model evaluation, and creating interactive ML applications with Streamlit and FastAPI. Proficient in backend API development using FastAPI, database management with PostgreSQL, and ORM integration using SQLAlchemy. Strong understanding of NumPy, Pandas, Matplotlib, Seaborn, and modern AI workflows, with a passion for building scalable, production-ready AI systems and continuously learning emerging technologies in Artificial Intelligence and Machine Learning.

Core Competencies

AI/ML Development: Machine Learning Model Development, Supervised & Unsupervised Learning, Classification & Clustering, Model Evaluation, Feature Engineering, Data Preprocessing

Machine Learning Frameworks & Libraries: Python, Scikit-learn, TensorFlow, PyTorch, NumPy, Pandas, Matplotlib, Seaborn

Natural Language Processing: Sentiment Analysis, Fake News Detection, Text Classification, TF-IDF Vectorization, NLP Pipelines

Computer Vision: Image Recognition Systems, Image Classification, Deep Learning Model Implementation

Backend & API Development: FastAPI, REST API Development, Backend Integration, Model Deployment, API Testing

Database Technologies: PostgreSQL, SQLAlchemy, SQL Database Design, Data Query Optimization

Data Analysis & Visualization: Exploratory Data Analysis (EDA), Statistical Analysis, Data Cleaning, Business Insights Generation

Deployment & Application Development: Streamlit, Interactive ML Applications, Model Integration, Production Deployment

AI System Optimization: Hyperparameter Tuning, Performance Evaluation, Model Validation, Handling Imbalanced Datasets (SMOTE)

Development Tools & Workflow: Git, GitHub, Jupyter Notebook, Google Colab, Version Control, Debugging & Documentation

Professional Experience

• **Intern at Developers Hub (Remote)** — ML Engineer 2025 – Islamabad, Pakistan

- Developed and deployed Machine Learning models for NLP, classification, and clustering projects using Python and Scikit-learn.
- Performed data preprocessing, feature engineering, and model evaluation to improve prediction accuracy and system performance.

- Built interactive AI applications using Streamlit and collaborated on real-world AI/ML solutions following industry best practices
- **Intern at Code Alpha (Remote) — ML Engineer 2025 – India**
 - Developed and deployed Machine Learning models for NLP, classification, and clustering projects using Python and Scikit-learn.
 - Performed data preprocessing, feature engineering, and model evaluation to improve prediction accuracy and system performance.
 - Built interactive AI applications using Streamlit and collaborated on real-world AI/ML solutions following industry best practices.

KEY PROJECTS

Fake News Detection System

Developed an NLP-based fake news detection system using TF-IDF vectorization and Machine Learning algorithms to classify news articles with high prediction accuracy.

Customer Segmentation Analysis

Built a customer segmentation model using K-Means clustering to identify customer groups and generate business insights through data visualization and analysis.

Movie Review Sentiment Analysis

Designed a sentiment analysis system using Natural Language Processing and Logistic Regression to classify movie reviews as positive or negative.

Credit Risk Analysis System

Developed a predictive credit risk model using financial datasets, feature engineering, and SMOTE techniques to identify high-risk customers and improve classification performance.

Image Recognition System

Implemented a deep learning-based image recognition system for image classification tasks using Python and computer vision techniques.

AI/ML Web Applications with Streamlit

Created interactive Machine Learning web applications using Streamlit for real-time predictions, model

visualization, and user-friendly AI deployment.

FastAPI-Based ML Backend APIs

Built RESTful APIs for Machine Learning models using FastAPI, integrating PostgreSQL and SQLAlchemy for efficient backend and database management.

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Technical Skills

Programming Languages: Python

Machine Learning: Supervised Learning, Unsupervised Learning, Classification, Regression, Clustering (K-Means), Model Evaluation, Feature Engineering

Natural Language Processing (NLP): Sentiment Analysis, Fake News Detection, Text Preprocessing, TF-IDF Vectorization

Deep Learning & Computer Vision: Image Recognition, Image Classification, Neural Network Basics

Libraries & Frameworks: Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn

Backend Development: FastAPI, REST API Development, Model Deployment

Databases: PostgreSQL, SQL, SQLAlchemy

Data Analysis: Exploratory Data Analysis (EDA), Data Cleaning, Data Visualization

Deployment Tools: Streamlit, Jupyter Notebook, Google Colab

Version Control: Git, GitHub

Education

Bachelor of Science in Software Engineering Islamic University Bahawalpur 2023 – 2027

- Specialization in AI/ML

FSc Pre-Engineering Kips College 2021 – 2023

Certifications & Professional Development

Programming Fundamentals in C++ and DSA — IUB

Python Crash Course — Programming with Mosh (YouTube)

Intro to Machine Learning — Codanics (YouTube)

Machine Learning Algorithms — Sidarthan (YouTube)

Flutter 2022 BootCamp — Tech Brothers Asif Taj

Languages

English (Professional Working Proficiency)